

$$y_{n+1} = \frac{y_n + \frac{x}{y_n}}{2}$$

In this case, the mathematician would be relying on the common mathematical practice of describing infinite sequences by continued subscripts, as in: $(y_1, y_2, \dots, y_n, \dots)$. If you know the n th term of this infinite sequence, you can get the $n + 1$ th term from the equation above. This sequence will converge to $\sqrt{x}$, or in other words $y_\infty = \sqrt{x}$. Whether you prefer the clarity of the four-step description, the elegant concision of an equation, or the practicality of the code we will write is a matter of taste, but it helps to be familiar with all the possible ways to describe an algorithm.

You can understand why the Babylonian algorithm works if you consider these two simple cases:

- If $y < \sqrt{x}$, then $y^2 < x$. So $\frac{x}{y^2} > 1$ so $x \times \frac{x}{y^2} > x$.

But notice that $x \times \frac{x}{y^2} = \frac{x^2}{y^2} = \left(\frac{x}{y}\right)^2 = z^2$. So $z^2 > x$. **This means that $z > \sqrt{x}$.**

- If $y > \sqrt{x}$, then $y^2 > x$. So $\frac{x}{y^2} < 1$, so $x \times \frac{x}{y^2} < x$.

But notice that $x \times \frac{x}{y^2} = \frac{x^2}{y^2} = \left(\frac{x}{y}\right)^2 = z^2$. So $z^2 < x$. **This means that $z < \sqrt{x}$.**

We can write these cases more succinctly by removing some text:

- If $y < \sqrt{x}$, then $z > \sqrt{x}$.
- If $y > \sqrt{x}$, then $z < \sqrt{x}$.

If y is an underestimate for the correct value of $\sqrt{x}$, then z is an overestimate. If y is an overestimate for the correct value of $\sqrt{x}$, then z is an